

2025 DRP Talk: Can everything be untied?

Sammith Belur and Havish Goutham Prageeth

Department of Mathematics
Purdue University

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What is Knot Theory?

- To understand what knot theory is, we need to look at the mathematical definition of a “knot”.
- Physically, knots are usually seen as tangles made in bits of string/rope or something similar.
- However, mathematically:
 - Knots are closed loops
 - Is an embedding of the circle S^1 into $\mathbb{R}^3$



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1. Introduction

2. Motivation

3. Reidemeister
Moves

4. Knot Invariants

5. Unsolved
Problems

How can Knots be represented?

- A knot diagram is a 2D projection of a 3D knot with over/under crossing information
- The same Knot can have multiple diagrams (will be talked about later!)
- Computationally, Knots can also be visualized using Mathematica with the KnotData package or the more advanced KnotTheory package

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2D Projection Example

a



→
Projection



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Types of Knots

- The most basic knot is called **The Unknot (aka Trivial Knot)**.
- The most basic non-trivial knot is the **Trefoil**
- Next simplest knot is the **Figure-Eight Knot**
- Beyond these, there are infinitely many distinct knots, all with unique properties and structures
 - Some examples: Pretzel Knot and Torus Knot among several others

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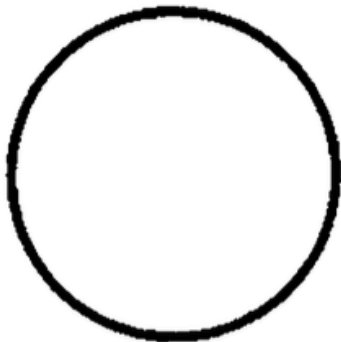
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The Unknot



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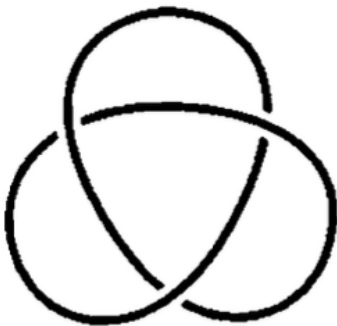
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The Trefoil



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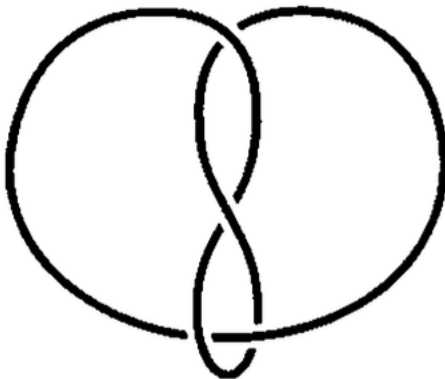
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Figure-Eight Knot



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Why study Knot Theory?

- Appears in DNA loops and Molecular Biology
- Provides insight into Molecular Structure in Chemistry
- Fundamental to low-dimensional topology
- Applications in Quantum Computing

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Can every knot be untangled?

Two questions can be thought about to answer this:

- When are two knots considered "the same"?
- What changes can we make to the 2D representation of a knot that don't alter the knot?

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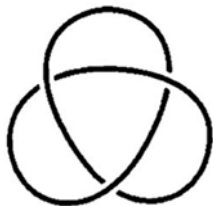
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Are these two Knots the same?



What can we say about this knot?



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Reidemeister Moves

- Two knot diagrams represent the same knot if and only if they are related by **Reidemeister moves**
 - The “twist”
 - The “finger move”
 - The “crossing slide”
- Under Reidemeister Moves, even if a knot looks different, its properties are the same
 - We can use Reidemeister moves to show that one knot is equivalent to another

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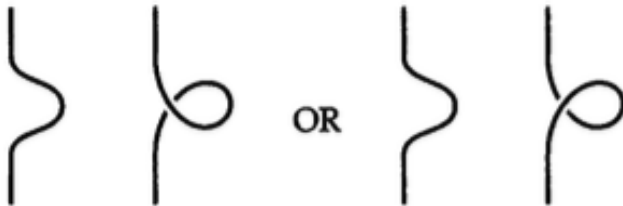
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Type I Reidemeister Move (The “twist”)



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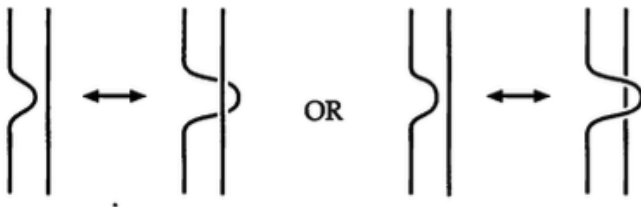
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Moves**

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Type II Reidemeister Move (The “finger move”)



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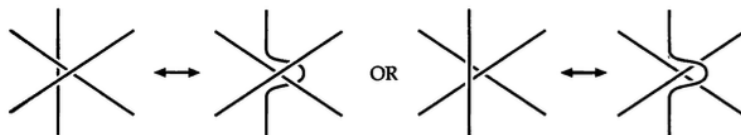
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Type III Reidemeister Move (The “Crossing Slide”)



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Knot Invariants

- A **Knot Invariant** is a property of a knot that remains the same under Reidemeister Moves
- A few types of Knot Invariants:
 - Crossing Number
 - Tricolorability
 - Knot Polynomials
- In this talk, we will focus on Tricolorability

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Tricolorability

- A knot is tricolorable if you can color its strands in some projection using **three** colors such that:
 - At every crossing, all three strands have different colors, or all three are the same color
 - At least two colors are used overall
- It remains the same under Reidemeister moves, so it is an invariant!

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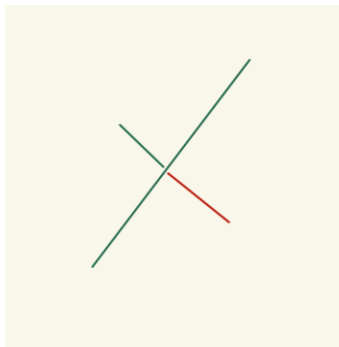
2. Motivation

3. Manipulation

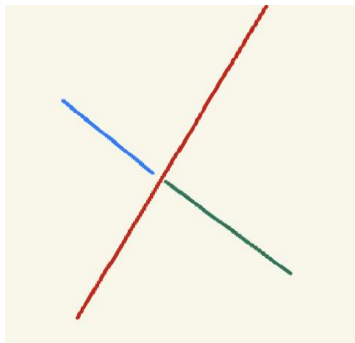
4. **Knot Invariants**

5. Unsolved
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Violates conditions
of Tricolorability



Does not violate
conditions of
Tricolorability



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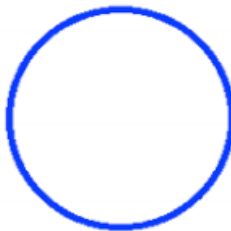
5. Unsolved
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Tricolorability Examples:

The Trefoil is tricolorable



The Unknot is not tricolorable



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Some Unsolved Problems in Knot Theory

- Write a computer program that can tell if a given knot is the unknot. No known efficient algorithm currently exists.
- Among the several invariants that exist, no known knot invariant can fully distinguish all knots
- All knots up to 16 crossings have been classified, there are no efficient methods known to classify knots at higher complexity

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To answer our question in the title....

Not everything can be untied.

Thank you everybody for attending!

Special thanks to our mentor Nicolas Bridges for his constant support and the organizers of the DRP for providing us with this opportunity.

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References:

- The Knot Book – Colin Adams
- Knot Theory – Shintaro Fushida-Hardy
- Figures in the slides are drawn from The Knot Book – Colin Adams.
- Aaron Cheng – Not all knots are projections of the Unknot
- Composite Knot Determinants